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Saffron cultivation using aeroponics: A promising approach for viable and high-quality production

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Abstract

The method of growing plants without the utilization of soil is known as hydroculture**.** There are three hydroculture techniques such as aeroponics, hydroponics, and aquaponics. The word aeroponics derives from the Latin word '*aero*' which means air and '*ponic*' which means labour /work. In aeroponics, plants dangle in the air and receive nutrients from a mist-fed nutrient solution. Saffron also known as "orange gold," holds a distinguished status among spices for its rarity, valuable properties, and versatile applications in both medicine and religious ceremonies. India has the 2nd largest area under saffron cultivation. The main concern with saffron cultivation is low productivity and specific climate requirements, which can be solved by using the aeroponics technique. The aeroponics techniques provide several advantages over conventional agriculture like high space efficiency, meeting climatic requirements that enable year-round production. Under aeroponics, the root length of saffron plant increases and simulate the flower initiation due to combine application of ɤ-aminobutyric acid with blue and red lights. In India, the future scope of saffron cultivation under aeroponics holds several potential benefits such as geographical expansions, maintaining climate control, conserve water, producing standardized-grade saffron and improving export capacity, which will potentially increase the share of Indian agriculture in the country's GDP.

Keywords: Hydroculture, aeroponics, saffron, hydroponics, aquaponics, GDP, orange gold, ɤ-aminobutyric acid

1. Introduction

In the Indian economy, agriculture is considered the backbone, contributing around 18% of the gross domestic product (GDP) and 43% of the geographical area (Dolli & Divya, 2020) [6]. As the population is increasing, the need for food is also increasing, and to meet that requirement, we need continuous development of new technologies. Hence, there is a need to maximize the economic potential of agriculture (Gubbi *et al*., 2013; Alexandratos and Bruinsma, 2012) [26, 27]. It can be achieved by improving the cultivation of economically potent crops like saffron (*Crocus sativus* L.). Herat is the world's largest producer which contributes 90% of the world's production. However, in some countries, there is a marginal fall in the production of saffron. The use of IoT sensors (Internet of Things) can be a great solution to this problem and it is also used in various smart agriculture to monitor environmental conditions that ensure high saffron production with great quality (Perera *et al*., 2014; Jin *et al*., 2014) [28, 29]. Aeroponics (The Latin word '*aero*' means air and '*ponic*' means labour /work) is one of the most widely used technologies for the artificial growth of crops. Most of the problems in traditional agriculture like increased crop duration, space requirements, lack of nutrients, and high labour requirements may readily be solved by artificial production methods *(*Kumari & Kumar, 2019) [13]. The method of growing plants without the utilization of soil is known as hydroculture. This involves growing plants in any medium other than soil such as rock wool, clay pebbles, or aquatic medium (Zsombik *et al.,* 2021) [25]. There are three hydroculture techniques such as aeroponics, hydroponics, and aquaponics. Like other hydroculture techniques, in aeroponics, plants dangle in the air, but aquaponics makes use of the entire ecosystem so that the output of aquatic creatures may contribute to plant growth. The space requirement, cost, the capacity to grow certain crops, and, the possibility of a single point of failure that would cause the entire ecosystem to collapse are the problems with using any of these methods (Çalı̧skan *et al.,* 2021 & Bhakar *et al.,* 2021) [4, 3]. In aeroponics, the main concern is producing a high-quality saffron stigma (*Crocus sativus* L.).

Saffron also known as "red gold" (Sut *et al*., 2024) [21], holds distinguished status among spices for its rarity, valuable properties, and versatile applications in both medicine and religious ceremonies. The glucoside which is responsible for the colouring of saffron is crocin and the bitter substance is picrocrocin, the other main constituents are crocetin and safranal. As it has some medicinal properties, it is used to treat melancholia, fever, liver, and spleen enlargement etc. It also has stimulant and stomachic properties and is a remedy for catarrhal affection in children. In India, it is an important ingredient for both Ayurvedic and Unani systems of medicine. Other than this it is also used in flavoring and colouring dishes. There are some indigenous varieties (SKU-S-86-KIS, SKU-S-86 WAT, SKU-S-86-QUIL, SKU-S-86-Sona, SKU-S-86 Lath, SKU-S-86 Chand, SKU-S-86 Chad), and some exotic varieties are EC-37328, EC-221039, EC-217010, SKUAST, Nag C8708 and Badi C8606. To increase the production, aeroponics is adopted.

The principal saffron-growing countries are Iran, India, Greece, Morocco, Spain, Italy, Turkey, France, Switzerland, Pakistan, China, Japan, and Australia. Iran is the world's top producer of saffron, meeting about 80% of the global demand (Menia *et al., 2018*) [14]. India has the 2nd largest area under saffron cultivation, but accounts for 7% of the world's total production. J&K holds 3715 ha for production, and productivity accounts for 3 - 4 kg/ha. It is largely cultivated in four districts (Pulwama, Budgam, Srinagar & Kisthwar), with 86% of that crop grown on 3200 ha in Pampore, a historic site. Saffron productivity and area decreased by 83% and 72%, respectively, between 1997 and 2015 (Husaini *et al., 2010*) [9].

2. Major Problems related to the production of good quality saffron in India

Various studies on saffron cultivation have identified several problems related to saffron production in India. So, these problems are either directly or indirectly linked to the development of the saffron industry.

2.1 Improper Marketing Strategies

The saffron industry in Kashmir is mainly controlled by a few individuals, as small growers are unable to grade, pack, and store the saffron on their own (Zaki, 2002) [24]. From farmers to the patron, these few individuals are connected. So, this leads to the establishment of several middlemen and intermediates who obtain the lion's share of economic benefits from saffron farming. Regional merchants, brokers, and cooperative organizations are included in marketing channels. The middlemen who are connected with farmers and patrons take a sizable profit to home, as the saffron market in Kashmir is unorganized. Nearly 71% of saffron is sold to brokers and only 16% and 13% of saffron is sold through subsidiary businessmen and other agencies. As a result, farmers who sell their products through subsidiary businessmen and other agencies face drastic exploitation.

2.2 Mismanagement in Harvesting

In India, there are some post-harvesting issues, that result in low-quality production of saffron. There is a decrement in crocin caused by a delay in harvesting during the opening and a delay in stigma separation from the flower. This mismanagement of harvesting and improper packaging results in a negative effect on the standard of saffron and, subsequently a reduction in the farmers' economy.

2.3 Mixing of impurities

Mixing synthetic adulterants is also reducing the quality of saffron. Lack of management results in the smuggling of cheap saffron from Iran and then mixing it with these, reducing the standard. These unethical practices harm the saffron industry and the reputation for high-quality saffron is also significantly damaged.

To bring the efficiency of the saffron market, this asymmetrical market needs to be recognized as a severe issue. Government should pay attention to adulteration which is a serious problem in degrading saffron quality. To prevent these issues at the national and international levels, practices like IoT-based technology such as aeroponics are required.

3. Aeroponics as a Vertical Farming

Aeroponics is also known as vertical farming due to its distinct vertical structure.

3.1 Maximizing Space Efficiency

Aeroponics can vertically stack multiple layers of plants, allowing a large number of plants to be produced in a compact area. The traditional farming method depends on large horizontal expanses; however, vertical farming utilizes vertical space, making it far more space-efficient and increasing productivity in a limited area.

3.2 Overcoming Spatial Constraints

Vertical farming is particularly beneficial in urban areas where land is scarce or expensive. Farms may be developed in high-rise buildings or repurposed structures using aeroponics, allowing for localized food production without requiring excessive space. This vertical crop cultivation system can address the space and land shortage concerns that many metropolitan regions confront.

3.3 Enhanced Resource Manage

Vertical farming through aeroponics provides a possibility to maximize resource efficiency. Farmers can optimize resource use by cultivating plants in a controlled environment. The mist-based aeroponic technology delivers nutrients directly to the plant roots, reducing waste and increasing nutrient uptake. Additionally, regulated settings limit the necessity for pesticides or herbicides, resulting in more sustainable and environmentally friendly farming practices.

3.4 Year-round Production

Vertical farming enables year-round crop cultivation by eliminating the constraints imposed by seasonal changes. The regulated environment of aeroponic systems provides compatible temperature, light, and nutrient availability, allowing farmers to grow crops all year. This ongoing production capability is especially beneficial in areas with harsh temperatures or short growing seasons, assuring a consistent and dependable food supply.

3.5 Technological Integration

Vertical farming relies largely on advanced technologies like LED lighting, artificial systems, and automation systems to ensure optimal growing conditions. These technologies allow for precise control of environmental parameters like humidity, temperature, light intensity, and nutrient distribution. Vertical farming separates itself from traditional agricultural operations by the synergistic combination of technology and aeroponics.

4. Working

There is a good monitoring and control system required for the delivery of water and nutrients to corms and root systems. The distribution of fertilizer solution is done by pipes, spray nozzles, a pump, and a timer. The nutrient-rich solution is sprayed on the roots of saffron with the help of an internal microjet spray which creates high pressure with fine particles.
The root hairs help to take nutrients from moisture. Even the wastage of nutrients is very tiny in amount. The misting occurs every several minutes around the dangling roots. Normally, the mechanism would activate for a few seconds every two to three minutes. The roots will soon dry up since they are exposed to the air if the misting cycle is interrupted. The nutrient pump, like in conventional hydroponics systems, is controlled by a timer; however, aeroponics requires a short cycle timer to run the pump for a brief period every few minutes. The chamber must be filled with lightless materials to keep the roots functional and avoid algae development, which impedes plant growth and pollutes the system. As large droplets may decrease the quantity of oxygen, on the other hand, undersized droplets may reduce the effectiveness of an aeroponic system by promoting root hair development while suppressing lateral root growth for this reason droplet size is important. The droplets must be adequately sized to transport nutrients to the roots. The unused liquid that leaks from the unit's base is strained, filtered, and pumped back into the reservoir (Kumari & Kumar, 2019) [13]. Aeroponics allows simple pH and nutrient monitoring, supporting plant growth with minimal contact with a support system.

5. Growth of Saffron Under Aeroponics

Under aeroponic conditions, an increase in dry weight of corms has been seen. The number of roots is increased in aeroponics, but their length is significantly reduced, resulting in shorter, thicker, curlier, and more contractile roots. In aeroponics, root growth is limited to a certain length, which corresponds to the depth of the growing chamber. The shoot appearance of saffron under aeroponics is much earlier than soil-cultured plants (Souret and Weathers, 2000) [20]. The colours of stigma harvested from aeroponics are identical to soil-cultured saffron. The crocin and crocetin are higher in aeroponics stigmas but picrocrocin concentrations are lower and there is no change in safranal concentration. Some factors influencing the flowering of the bulb such as temperature, relative humidity, storage time, bulb size, soil temperature, water stress, light intensity, defoliation, etc. Saffron transplanting and corm production in aeroponics increases the corm size but it increases the production of contractile roots and drying of aerial parts has been seen. Quantification of safranal and picrocrocin in saffron stigmas using PAFT improves the quality of saffron. The highest concentration of mineral nutrients produced the finest quality corms, resulting in a three to five-times higher yield of corms larger than 25 mm in diameter. An EC of 2.5 $dS \cdot m^{-2}$ resulted in the maximum number of $corms \cdot m^{-2}$ and a 20% increase in corm yield. Increasing nutrient content in fertigation solution led to a considerable increase in nutrient uptake (Salas *et al*., 2000) [18].

La^{3+} (60 µM) significantly promoted callus development but only marginally raised crocin content.
Crocin content was considerably increased by Ce^{3+} (30 µM) but cell growth was not significantly impacted. The simultaneous promotion of cell proliferation and crocin content may in a synergistic effect when La^{3+} (60 µM) and Ce^{3+} (20 µM) are added (Chen *et al*., 2004) [5].
The application of plant growth regulators and different light conditions improved the flower production of saffron. Application of ɤ-aminobutyric acid with blue and red lights combination increases the number of flowers per corm, and the stigma fresh and dry weight of saffron. Also significant increase in the fresh weight of stigma has been seen in the case of ɤ-aminobutyric acid and red light combination. The increase in the amount of safranal and picrocrocin is due to ɤ-aminobutyric acid and red and blue light combination. The crocin content is more in stigmas of corms due to gibberellic acid and blue and red-light effects (Eftekhari *et al*., 2023) [7].
Using a six-layer vertical system reduces flower production and increases the saffron yield by up to 19 times compared to soil culture. Using a one-layer hydroponic system increases yield by up to four times. The two hydroponic systems had three times plant density per layer compared to soil, which likely impacted vegetative development and corm yield (A. Soqeer *et al*., 2024) [19].
The optimal climatic conditions, particularly a favorable temperature during the flowering emergence stage, cause a significantly higher proportion of flowering compared to conventional production techniques (Koocheki *et al.,* 2020) [10]. During the 2018 and 2019 flowering seasons, Shajari *et al.* assessed how environmental factors and production practices affected saffron yield and quality. The study examined the differences between planting corms in soilless and conventional soil production methods. For saffron grown in the two systems over both During growing seasons, the results showed significant differences in all flowering traits. Only 6% of the corms planted under traditional production produced flowers during the first season, when planting densities were maintained at the same levels for both methods; in contrast, 38.6% of corms planted using soilless production produced flowers. The traditional production technique had a slightly lower blooming percentage of 56% in the second season, compared to 65% with the soilless production system (Shajari *et al*., 2021) [1].
In an indoor enviornment, the density of saffron cultivation may be maximised. For example, up to 4 t of saffron bulbs may be cultivated in a 60 m < sup > 2 < /sup > hall. 1 kg of dried saffron is Produced for every 70 kg of collected wet flowers, resulting in one tonne of saffron bulbs. produces 2 kg of dried material. As a result, one hectare of outdoor saffron growing may yield the same quantity of saffron as 100 meters < sup > 2 < /sup > of indoor space (Nelson, 2021). With control solution, saffron crops that get 30% potassium fertigation are more productive and of high quality (Valenzuela *et al*., 2024). Increasing nitrate concentration up to 9800 µM in the nutrient solution enhances leaf weight, photosynthesis rate, and the proportion of large cormlets (>8 g). However, higher iron concentrations can reduce the weight of both total and large cormlets, while boron concentration shows minimal effect on growth parameters (Naseri *et al.*, 2025) [16].

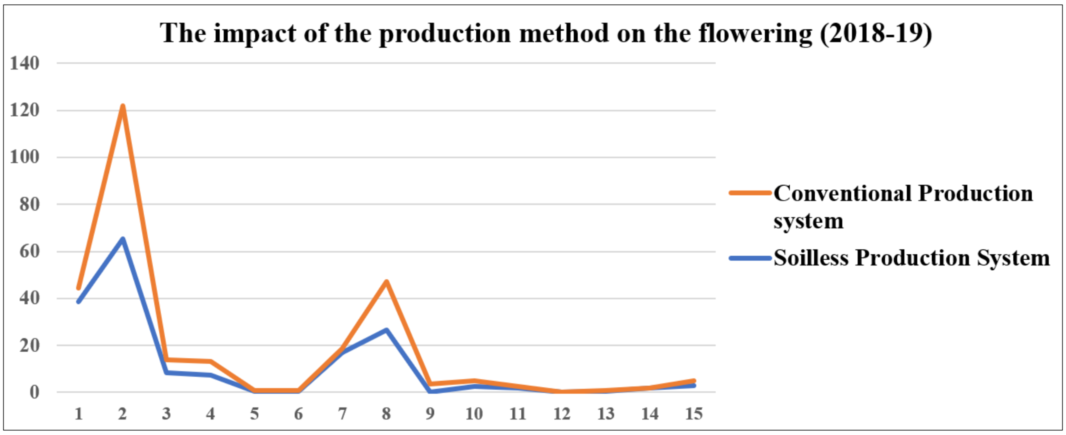

Fig 1: The impact of the production method on the flowering

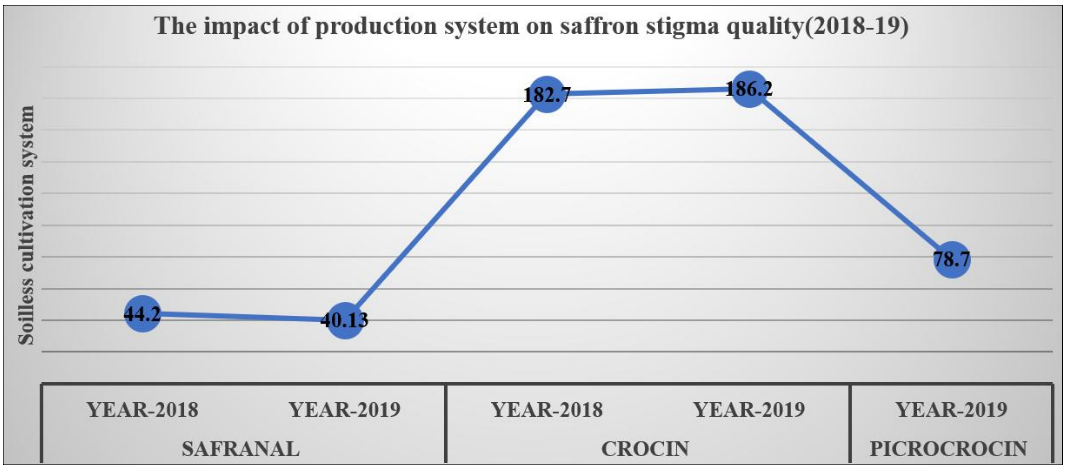

Fig 2: The impact of production system on saffron stigma quality under soilless cultivation

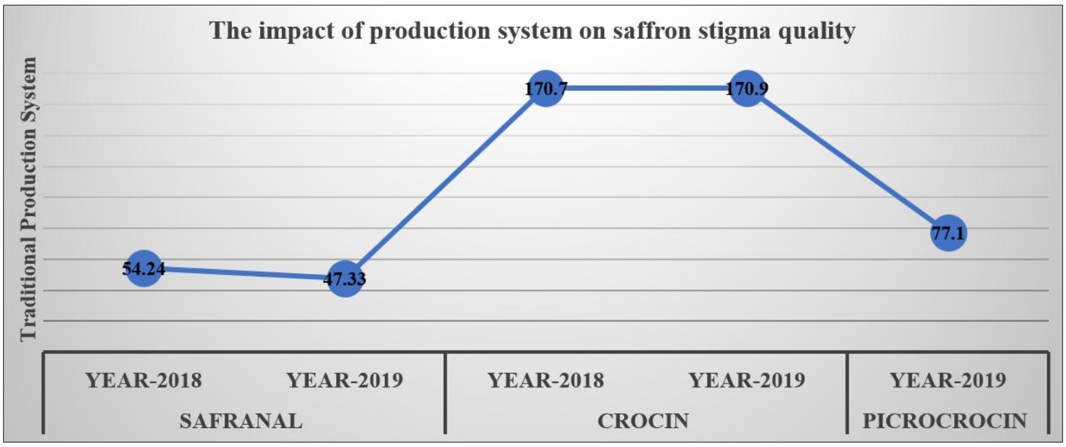

Fig 3: The impact of production system on saffron stigma quality under traditional cultivation

6. Future Scope

In India, the future scope of saffron cultivation under aeroponics holds several potential benefits:

1. Saffron cultivation is limited to regions like Jammu and Kashmir, some parts of Uttarakhand, and Himachal Pradesh. With the help of aeroponics technology cultivation of saffron can be expanded to other non-traditional regions of India. This can lead to increased saffron production and economic growth in these areas.

1. The Indian climate is diversified with varying temperatures and rainfall patterns. Aeroponics controls the environmental factors and mitigates the risk associated with unpredictable weather conditions.
2. India faces significant water scarcity challenges, particularly in drought-prone regions. Under aeroponics water use efficiency is higher which helps in sustainable saffron cultivation in such areas.
3. Saffron produced under aeroponics can exhibit consistent quality attributes, including colour, aroma, and flavour. Through controlled environmental conditions, aeroponics offers the potential for standardized saffron production, meeting both domestic and international quality standards.
4. With the help of saffron cultivation aeroponics can

create employment opportunities, particularly in rural areas. The technology requires skilled labor for the installation and maintenance of aeroponic systems, and the increased production potential can lead to additional jobs in saffron processing, packaging, and marketing. This can provide income-generation opportunities for rural communities and contribute to local economic development.

5. The demand for saffron, both domestically and internationally, is growing. Saffron cultivation with aeroponics can boost India’s export potential by ensuring constant quality, higher production, and the capacity to fulfill year-round production. This can improve India's standing in the global saffron market to boost foreign exchange profits, which will potentially increase the share of Indian agriculture in the country’s GDP.

Overall, saffron cultivation with aeroponics has a promising future in India. The capacity of aeroponics to overcome geographical constraints, maintain climate control, conserve water, and produce standardized-grade saffron has a promising future in India.

7. Conclusion

Aeroponics in saffron cultivation offers numerous benefits, including soil elimination, reduced disease risk, improved water and nutrient use efficiency, and precise control over environmental factors for crucial saffron growth and quality. Aeroponics contributes to higher saffron yield by optimal oxygen and nutrient supply, promoting faster growth rates and enhanced biomass production, enabling frequent cropping cycles and multiple harvests within a year. Aeroponics for saffron cultivation contributes to sustainability, resource conservation, and efficient use of agricultural inputs. However, precise monitoring and expertise are required for successful implementation. Continued research could revolutionize saffron production.

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